

## **Reviewer Instructions**

This document provides instructions for reviewers evaluating outputs generated by Large Language Models (LLMs) within the study titled:

### **“An Empirical Evaluation of LLM-Assisted Sketch-Based Requirements Elicitation and Prototyping”**

#### **Evaluation Materials**

- Original low-fidelity sketches
- Generated natural-language requirements
- Generated HTML/CSS/JavaScript prototypes
- Multiple independent runs for each model

#### **Evaluated Models**

- GPT-4.1
- GPT-4o
- Claude Sonnet 4.5
- Gemini 2.5 Pro

#### **Experimental Scenarios**

- E-commerce Checkout System
- Healthcare Appointment System
- Service Portal System

#### **Reviewer Tasks**

- Examine the provided sketches carefully
- Review the generated requirements
- Execute and inspect the generated HTML/CSS/JavaScript prototype locally
- Complete the scoring workbook
- Provide optional qualitative comments

#### **Prototype Evaluation**

- Visual Fidelity (VF)
- Functional Alignment (FA)
- Structural Code Quality (SCQ)
- Human-Perceived Usability (HPU)

#### **Requirements Evaluation**

- Semantic Accuracy (SA)
- Requirement Coverage (RC)

- Novelty Precision (NP)
- Clarity & Readability (CR)
- Consistency (CO)

### **Scoring Procedure**

All evaluation criteria use a 5-point Likert scale:

1 = Very Poor

2 = Weak

3 = Adequate

4 = Good

5 = Excellent

### **Important Notes**

- Please evaluate each run independently
- Focus on output quality rather than model reputation
- Maintain consistency across evaluations
- Optional comments are highly encouraged

Thank you for contributing to this evaluation study.